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Research article

Epistemic rhythms in lesson assignments: exploring the interplay of diverse content and objectives in teaching materials on historical and contemporary societal issues

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Abstract

This article examines the design of lesson assignments addressing historical and contemporary societal issues in schools, with particular emphasis on the knowledge content required to respond to the questions posed in these assignments. The aim is to investigate the didactic functions that can be identified in school lesson assignments that elicit various types of knowledge. Two research questions are posed: (1) What epistemic positions can be identified in school lesson assignments? (2) What epistemic rhythms between different selections of knowledge can be identified in school lesson assignments? The methodology used involves a text analysis of 124 lesson assignments provided by 10 upper secondary teachers. The results identify three primary epistemic rhythms in the lesson assignments, which are the movements between student-oriented, context-oriented and analysis-oriented knowledge. These correspond in different ways to the didactic functions of broadening, reorganisation and evaluation. An overarching conclusion is that teachers' approaches to designing lesson assignments with varying

epistemic rhythms can serve as a means to address diverse educational objectives and reflect teachers' practice-oriented professional theory. Therefore, this article argues that the concept of epistemic rhythms can serve as a robust framework for reflection when balancing knowledge and objectives in education. It asserts that the concept of epistemic rhythms reveals teachers' practice-oriented theories regarding the selection of knowledge for teaching and learning.

Keywords epistemic rhythm; lesson assignments; perspective of the teaching profession; historical and contemporary societal issues

Introduction

Research question

This article examines the design of lesson assignments concerning historical and contemporary societal issues within schools, with particular focus on the knowledge content required by the questions included in these assignments. A key premise of this article is that questions with varying knowledge content serve different purposes and objectives in teaching (Berg, 2014; Olsson, 2016). A recurring issue, from both contemporary and educational-historical perspectives, is how different considerations of knowledge content in school subjects contribute to various educational aims and objectives (Persson, 2017; Samuelsson, 2024). Furthermore, the role of school subjects in addressing contemporary societal challenges, such as climate crises, wars and conflicts, has been explored in relation to the question of knowledge in schools (Nordgren, 2021). Concurrently, discussions have also centred on the selection of knowledge that supports students' processes of meaning-making (Deldén, 2023).

The question of content and purpose within subjects like history addressing historical and contemporary societal issues has consequently attracted significant research interest (see Barton and Levstik, 2009; Berg, 2014; Lindmark, 2013; Odenstad, 2010). This has led to research across various areas. A particular interest has been directed at the governance of knowledge within schools. Specifically, the formulation and interpretation of curriculum documents in different subjects have been examined, with comparative studies highlighting changing epistemological orientations over time (Samuelsson, 2024). In both international and Swedish contexts, research has also focused on the content of national tests (Samuelsson and Wendell, 2016). Likewise, textbooks and questions within textbooks have been studied with similar findings over time, indicating that knowledge content has changed in relation to prevailing educational ideological positions (Nordgren, 2006; Persson, 2019).

Less attention has been given to the knowledge content in school lesson assignments. In particular, there is a lack of subject-specific didactic studies demonstrating how the design and selection of knowledge in lesson assignments can contribute to varying educational aims. This article focuses on the movement between different knowledge within lesson assignments. To identify this movement, the concept of *epistemic rhythm* is formulated and used for the analysis.

The first component of the concept – *epistemic* – is defined here as knowledge, in contrast to epistemology, which refers to theories about the nature of knowledge. Another way to express this is to speak of knowledge (epistemic) within a subject, such as knowledge represented in a specific lesson assignment (Åström Elmersjö, 2024). This may include knowledge formulated within academia and/or knowledge embedded in school subjects as articulated in the curriculum for a specific grade level at a particular point in time. When such selections are made for a course within a school subject, the school subjects also reflect different epistemologies or domains of knowledge, that is, theories of knowledge as mentioned (Berg, 2023). Thus, when teachers select knowledge content in relation to different courses, they often do so – consciously or unconsciously – with reference to one or several epistemological orientations (see Elmersjö and Zanazanian, 2024). It is also important to note that the selection of knowledge is inherently linked to power; someone holds the authority to define what constitutes adequate knowledge within a given domain. In the case of curricula, this authority may rest with the state, academic representatives or interest organisations (Samuelsson, 2008). This suggests that what is considered adequate knowledge within a domain is shaped by competing viewpoints.

For instance, research on various historical-didactic theories has demonstrated how different epistemic foundations have influenced the formulation of curricula in the subject of history (Englund, 1988). This also implies that teachers can be regarded as active agents in shaping specific selections of knowledge in relation to teaching, for example, when designing lesson assignments (see Berg, 2021; Nordgren, 2024).

The second component of the concept – *rhythm* – is defined here as the movement that can occur between different epistemic positions within a subject's content. The concept of rhythm in teaching is based on the idea that instruction follows a certain pattern, shaped by teachers and students within the framework of a didactic contract (see Brousseau, 1984). Among others, Högberg (2015, 2017) has demonstrated that teachers, in their pedagogical practice, must be prepared to shift between different pedagogical ideals, which, in response to the situated conditions of teaching, creates what he terms a certain *pedagogical rhythm*. In this article, the concept is used to explore the extent to which such rhythm can also be observed in the selection of knowledge when designing lesson assignments for instruction. Previous studies have sometimes framed this as a problem, suggesting that shifts between various epistemic positions may cause knowledge content to appear directionless (see Wagner, 2024). However, this article examines the assumption that such movement can instead be understood as a prerequisite for achieving the intended goals and purposes of teaching. The potential movement between different epistemic positions within a lesson assignment is examined to determine whether it has the capacity to generate a specific rhythm that fulfils various didactic functions. In this article, this particular movement between different aspects of knowledge is referred to as an *epistemic rhythm*.

The article is based on 124 collected lesson assignments from 10 different Swedish high school teachers in school subjects, addressing historical and contemporary societal issues. The overarching aim is to investigate what didactic functions can be identified in school lesson assignments. The purpose of the article is operationalised through two research questions:

RQ 1. What epistemic positions can be identified in school lesson assignments?
RQ 2. What epistemic rhythms between different selections of knowledge can be identified in school lesson assignments?

Previous subject-didactic research, particularly regarding historical and contemporary societal issues, has often been characterised by deductively formulated theoretical perspectives. These studies have primarily aimed to examine or explain teachers' selections (see a critical discussion of this in Berg and Persson, 2020; Persson and Berg, 2023, 2024; Thorp and Persson, 2020). This article, however, adopts a more elaborative approach. This approach can be likened to what Berg and Persson have previously termed the *perspective of the teaching profession* (see Berg, 2021; Berg and Persson, 2023; Persson, 2022; Persson and Berg, 2022). It is based on the assumption that teachers, through their professional judgement, make various selections and considerations of knowledge content in relation to the situational conditions of teaching formulation and practice-oriented theory (Berg and Persson, 2023; Loughran, 2019; Persson and Berg, 2022). The article therefore explores what this theoretical perspective can contribute to the understanding of the selection of epistemic rhythm and didactic functions in lesson assignments.

Previous research

Research on school lesson assignments is multifaceted, varying in the study objects considered interesting, methodological approaches and theoretical claims (see Berg and Persson, 2023; Holland, 2024; Joyce et al., 2018; Wagner, 2024). Broadly speaking, research has explored what a lesson assignment can be in relation to other practices within the school context, such as the relationship between pre-lesson assignments and assessment strategies (Ukoh and Onifade, 2019). It also addresses how technology can be used to design lesson assignments (Hemby et al., 2006; Rightmyer and Larson, 2010). Lesson assignments may also be interwoven with homework (Boles, 1999; Zohar et al., 1998), which concerns assignments that can be viewed as preparatory for upcoming instruction (Ewell and Rodgers, 2014). A particularly focused area is assignments described in terms of the 'flipped classroom', examining the balance between work conducted during and outside lesson time (Cheng et al., 2019).

This article focuses on three specific research directions. The first is research concerning different theoretical views on the design of lesson assignments and how research can contribute to school practice (see Mebert et al., 2020). This includes how design impacts the various types of knowledge assessed (Hutchings et al., 2016; Isomäki et al., 2023). A commonly discussed concept in this context is lesson

design, which highlights different principles for conducting lessons. These studies often aim to enhance teachers' ability to formulate lesson assignments and optimise teaching to achieve, as it is termed, better student outcomes (Matsumura et al., 2002). This may involve how to design instructions for better student performance on assignments and teaching quality (Joyce et al., 2018; Ovute and Ede, 2015; Roessger et al., 2018). There are also critical perspectives within this research on assignments that too narrowly seek specific results, advocating instead for a broader range of assignments that address multiple aspects of knowledge (Holland, 2024). This can also be seen in relation to other critical perspectives on effect studies and their influence on the school's mission and knowledge mandate. Here, a distinction is made between the goals of schooling and Bildung, or between a didactic and a curriculum tradition for the work of the school. Researchers argue that the school's mission needs to be seen within a broader knowledge mandate tied to the question of citizenship or students' meaning-making (Deldén, 2023). At the same time, it also concerns the view of teachers' roles and responsibilities (see Persson, 2017, 2019).

The second area of research explores the structure, referred to as the didactic question *how*, of the process within a teaching sequence. For example, Berg and Persson (2023) have demonstrated how teachers use different forms of knowledge to address students' questions during lesson discussions. In order to highlight teachers' shifting strategies within a teaching sequence, Högberg (2015), among others, introduces the concept of *rhythm*, and in this specific case the concept of *pedagogical rhythm*, to describe how various intentions interact to create a rhythm in teaching. He demonstrates, through discussions among teacher students, how different intentions in lesson conversations contribute to a flow that can be likened to a rhythm. A central aspect of this is the time dimension, meaning there is a movement over time in teaching (Högberg, 2015, 2017). Research also points to the significant degree of unpredictability in the teaching process and the necessity for teachers to be prepared to switch between different approaches. Thus, a rhythm can be both recurring and divergent (Högberg, 2017). When it comes to subject-specific aspects of various knowledge choices, Berg refers to this phenomenon as the 'epistemic pendulum' (Berg, 2023).

Third, a central aspect of this article is research on the selection of knowledge content in teaching and lessons, referred to as the question of *what*. There is a substantial body of research addressing a broad spectrum of issues, including the content of curricula and textbooks (Ammert, 2010; Nordgren, 2006). Particularly relevant to this article is Persson's (2019) research, which examines the knowledge orientation of assignments in textbooks. This educational historical study finds that the most common type of textbook question focuses on specific and closed responses. He then identifies analytical questions and, subsequently, student-involving questions. Over time, he notes a shift where analytical and student-involving questions become more common, while closed questions requesting specific answers seem to remain constant (Persson, 2019).

The content of examinations has also garnered considerable interest. Numerous studies are based on more cognitively oriented theories, focusing on the complexity of both questions and students' responses (see Isomäki et al., 2023; Lee and Shemilt, 2003; Odenstad, 2010; Rosenlund, 2023). This research also includes interest in national tests. Several studies have analysed the extent to which students meet the requirements of these tests, using subject-specific didactic theory to assess students' ability to employ subject-specific concepts in their analyses (Samuelsson and Wendell, 2016). Related studies investigate how students meet the demands of using source criticism in relation to today's information flow (Nygren, 2019). Often, these analyses are conducted with specific theoretical frameworks for what is considered adequate knowledge.

The question of selecting knowledge also pertains to epistemic positions in lesson assignments. For instance, epistemic positions concerning historical knowledge content have been analysed (see Elmersjö and Zanazanian, 2024). One way to examine the question of epistemic positions in relation to teachers' understanding and practice of history teaching has been to analyse their selection of knowledge through broader theoretical perspectives on history as an academic discipline (Maggioni, 2024). Researchers argue that teachers tend to be inconsistent, often describing them as 'wobbling' between different epistemic positions (Maggioni et al., 2009; Mierwald and Junius, 2022; Miguel-Revilla, 2022; Perry, 1970; Stoel et al., 2022). Some scholars view this as being problematic due to the shifting ontological foundations that underpin these positions (see Wagner, 2024). The reasons behind this inconsistency have been explored by various researchers. Among them, Ledman (2014) suggests that history teaching and learning itself can be seen as inconsistent due to the subject's weak classification and weak framing, making it difficult to establish epistemological clarity (see also Bernstein, 2000).

Alternatively, other researchers have shown, or raised the question, that this phenomenon may be understood within a broader perspective on teachers' professional logic (Åström Elmersjö, 2024; Maggioni et al., 2009; see also Elmersjö and Zanazanian, 2024; Stoel et al., 2022). In this context, the concept of epistemic fluency has been introduced to describe an alternative logic that may better capture how teachers navigate historical knowledge in practice within varied historical and cultural communities (Parkes, 2024). In this way, it is argued that shedding light on the issue of knowledge selection for teaching requires an examination of teachers' selection of knowledge in relation to the enacted teaching, in order to bring greater clarity (Elmersjö and Zanazanian, 2024). Other researchers (Berg, 2023; Berg and Persson, 2023) approach this movement, or fluency, in teachers' selection from a teacher-profession perspective. They argue that the movement between different epistemic positions forms the basis for a practical theory, where this movement is seen as a way to address various knowledge goals. They suggest that teaching is always situated within specific conditions, and therefore, the selection and balancing of knowledge must be understood in relation to these factors, including students' prior knowledge and curriculum guidelines (see also Nordgren, 2024). They therefore call for research that illuminates the theoretical logic behind these practice-based and situated knowledge selections (Berg and Persson, 2020; Persson and Berg, 2024).

Overall, there is a broad field of research on lesson assignments. Previous studies have primarily emphasised the presence of different epistemological positions in the selection of curriculum content across various subjects. What is notably lacking, and explicitly called for, in the research landscape are studies that examine the epistemic movement that may occur within individual lesson assignments, particularly those that analyse this movement using practice-oriented theories.

Theoretical framework

The premise of this article is that the teaching profession is characterised by recurring decisions made in daily practice (see Baumert and Kunter, 2013; Berg, 2021; Irisdotter Aldenmyr, 2020; Loughran, 2019; Nilsson and Loughran, 2011; Nordgren, 2024). This involves, for example, balancing the governance of the school with the instruction formulated for individual lessons and lesson assignments (Ball and Cohen, 1999). This means that teachers must consider a range of factors, such as students' prior experiences and knowledge, the duration of lessons and the structure of the timetable when making these decisions (Frelin, 2010). In addition, considerations are made in relation to educational philosophical perspectives and educational policy objectives (Berg, 2014; Shulman, 1986). Thus, teachers' work is linked to a complex set of goals, with various situational conditions requiring the specific lesson to be adapted to the prevailing circumstances (see Berg and Persson, 2023). This results in a significant degree of unpredictability in the teaching sequence, where didactic questions, for instance, must be continuously assessed in response to the dynamics of the teaching situation (Persson, 2022). One such area is the formation of lesson assignments (see Karlsson, 2011). The analysis is therefore based on the assumption that a lesson assignment is shaped in relation to a complex set of goals, where many considerations are made within the situated conditions. This means that a lesson assignment can potentially contain various epistemic ideals, thus following the logic of the subject's situated conditions within the school context (see Berg, 2023).

The article uses two criteria to define a lesson assignment in school. The first concerns the context and the didactic questions of where and when. In this case, it refers to the lesson assignment being designed for a specific school subject. It may be a task worked on during a single lesson or part of a lesson. It can also involve a task that spans several lessons (Boles, 1999; Ewell and Rodgers, 2014; Zohar et al., 1998). Thus, a lesson assignment forms the basis for a teaching sequence that extends over time. The second criterion relates to the didactic *what* question. Previous research has identified several criteria regarding the content of a lesson assignment. Among other things, they may be built around relevant social and historical issues. They should also be analytical and focused on disciplinary concepts (Sandahl, 2015; Seixas and Morton, 2013).

A lesson assignment thus has the possibility, like assessments, to include one or more questions addressing various knowledge content and objectives outlined in the school's governance documents, such as curricula and subject plans (see Rosenlund, 2023). Consequently, a lesson assignment potentially contains, during a teaching sequence, a form of movement where different types of knowledge are solicited through the questions (see Berg and Persson, 2023). To make this movement visible, the entry and exit points of the lesson assignment are studied in this article. What questions are posed at the entry

point and what questions are posed at the end of the assignment? This potential knowledge movement within the lesson assignment between different questions is referred to, as mentioned in the introduction, as epistemic rhythm.

As noted in the introduction, *epistemic*, as the first component in the concept, is defined as a domain of knowledge that determines what is considered adequate knowledge within a particular area (VanSledright and Reddy, 2014). An aspect of this is that there may also be competing viewpoints within and between different domains of knowledge. In this case, it concerns which types of knowledge are solicited within the framework of a lesson assignment (see Berg, 2023). Additionally, the concept of rhythm, also defined in the introduction, reveals the movement that can potentially exist within a lesson assignment between different knowledge contents (see Högberg, 2015, 2017). This means that the movement between various knowledge aspects in a lesson assignment is given a specific pattern – an epistemic rhythm in the movement between the assignment's entry and exit points (see Berg and Persson, 2023).

The epistemic rhythm of the lesson assignments is tested against two different perspectives on the view of knowledge in school subjects. First, it is examined in relation to three different positions identified in previous studies of teachers' selection of knowledge in subjects such as history (see Berg, 2014). This premise represents a significant stance adopted in the article. In previous research, broader theoretical perspectives on subject knowledge, such as theoretical perspectives on history as an academic discipline, have typically served as the analytical framework for interpreting empirical material (see Maggioni, 2024). In contrast, the present analysis draws on practice-oriented theoretical categories. These are closely related categories that have been identified in a range of school subjects, including civics and Swedish (Hultin, 2006; Lindmark, 2013; Odenstad, 2010). In this study, these categories are explored in greater depth through an examination of their potential interrelationships within the context of a specific classroom task. The purpose is not to assess whether any of the categories are more significant or dominant, but rather to investigate how they may potentially relate to one another.

These positions – here termed contextual, analytical and student-oriented – have been conceptualised as ideal types in previous studies, forming a field of tension and movement that illustrates how knowledge can be described from a practice-oriented perspective (Berg, 2014). They involve context-oriented knowledge, which is akin to first-order concepts or content knowledge. This is based on terms tied to specific themes within the subject. For instance, in history, it could be a term such as serf; in social studies, minister; and in religious studies, priest. Additionally, there is analysis-oriented knowledge, referred to as second-order concepts or procedural knowledge, which includes terms such as cause and effect, centre and periphery, and source criticism. Finally, there is student-oriented knowledge, where students are involved in the question, requiring them to draw their own conclusions or imagine a scenario.

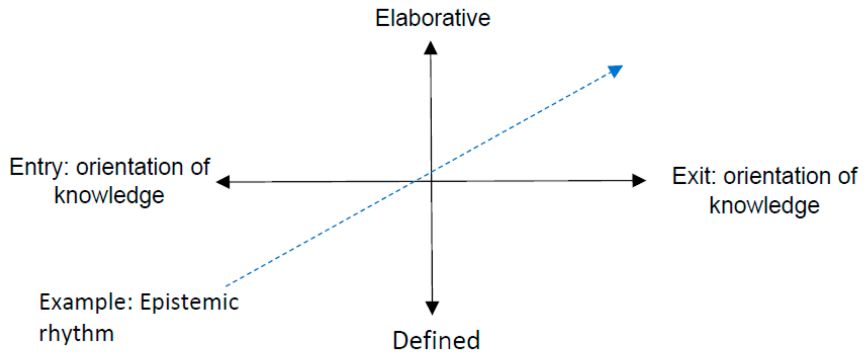
Second, the questions in the lesson assignments will be tested against two positions discussed within the field of historical didactics over the past decade (see Berg, 2023; Hammarlund, 2012; Persson, 2017). The first position focuses on the extent to which knowledge content can or should be clearly defined, emphasising the use of disciplinary concepts from academic subjects as a basis for schooling (Lee, 2005; Lilliestam, 2013). The second position concerns the extent to which school knowledge content should relate to a more open stance. For example, to what degree should the questions addressed in teaching be based on an elaborative approach, where there is no predetermined answer but rather an investigative character (Berg, 2021; Persson, 2019; Thorp and Persson, 2020)?

The content of a lesson assignment is thus viewed as a movement, as illustrated in Figure 1, between the knowledge orientation at the assignment's entry and its exit. It is also possible to identify the extent to which the lesson assignment's questions solicit knowledge that is clearly defined or more elaborative. The movement that can potentially be observed is illustrated in Figure 1 by the dashed blue line as an epistemic rhythm. It is assumed here that the specific epistemic rhythm observable in different lesson assignments can contribute to various didactic functions (see Berg and Persson, 2023).

If the entry of a lesson assignment, for example, is characterised by questions with a context-oriented and defined focus, and the exit features an analysis-oriented, elaborative focus, a specific didactic function can thus be observed. This would similarly apply to other variations. The question of didactic functions within teaching content has been examined in a number of studies. These include investigations into how teachers address students with different knowledge content in various teaching situations. The definition of didactic function in previous studies has been that the selection of knowledge content in a teaching sequence can contribute to various didactic functions. These have

included, among others, recreating, problematising, correcting, engaging, broadening and involving. In this way, it can be viewed as a tool to contribute to aspects of teaching that the teacher consciously or unconsciously aims to achieve. In this article, the didactic functions to which different selections of knowledge contribute remains an open question. These will be assessed inductively in relation to potentially identified epistemic rhythms in the lesson assignments (Berg, 2021, 2023; Berg and Persson, 2023; Persson, 2022).

Figure 1. Analytical model of epistemic rhythm in classroom assignments



Method

This article is based on a textual analysis of collected lesson assignments from school subjects addressing historical and contemporary societal issues. The subjects included as representatives of this category are history, social studies, religious studies and geography. A total of 124 lesson assignments were collected from 10 Swedish teachers who were qualified in, and taught, at least one of the subjects included in the study. The teachers were contacted through various digital platforms with an invitation to contribute materials for a study on lesson assignments. The teachers were affiliated with five different schools located in central Sweden. Teachers who expressed interest were asked to contact the researcher for further information about the study and to receive materials for informed consent. Following this, the teachers submitted a selection of lesson assignments of their own choosing and a signed blanket form for informed consent to participate in the study.

The materials were distributed among teachers and subjects, as outlined in Table 1. The article employs a variation theory approach for analysis. This means that the analysis focuses on variations in outcomes within the collected data. It is based on the assumption that, for example, a school assignment can be approached in different ways, and it is this variation that is of interest in the article (Marton and Booth, 1997). However, in seeking variation in this case, the intention is not to draw quantitative conclusions, but rather to identify patterns of variation. Instead, the analysis is based on the idea of identifying different variations of epistemic rhythms in the analysed lesson assignments. Furthermore, there is no intent to focus on individual teachers' lesson assignments as such in the analysis; rather, the assignments are treated as a collective group, where the variation is examined.

Table 1. Teacher codes and lesson assignments

Teacher	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Total
Number of lesson assignments	17	10	4	12	18	24	4	2	14	19	124

The analysis of the lesson assignments was conducted in three stages. In the first stage, all the lesson assignments were read through without making any specific analytical notes. This was done to gain an overview of the material. In the second stage, the analysis of the entries and exits in the assignments was conducted. The entry point consists of the initial question or questions in the assignment which,

individually or collectively, present a coherent body of knowledge. Similarly, the exit point is defined by the final question or questions in the assignment that together represent a coherent body of knowledge. The analysis of the entry and exit points was guided by the criteria previously defined in the theoretical framework. It focused on whether the questions were oriented towards student-, context- or analysis-based knowledge, and whether the knowledge being sought was elaborative or defined. These criteria were used to make visible the potential variation in how different types of knowledge were related within the assignments. They are more closely clarified later in the description of the analytical procedure.

On this basis, the questions at the entry and exit points of the lesson assignments were identified and highlighted. Subsequently, an assessment was made to determine whether the introduction was oriented towards student-, context- or analysis-oriented knowledge. Student-oriented knowledge refers to questions where the student is personally engaged with the issue, for instance, asking about their personal opinion on a topic (see Persson, 2019). Context-oriented knowledge refers to knowledge that is considered first-order concepts, that is, concepts used to describe collective issues and phenomena (see Lilliestam, 2013). In the context of social studies, this might involve the terms used to describe democratic institutions in society, such as parliament, mandates and government. Finally, we have analysis-oriented knowledge, which is described as second-order concepts (see Lee, 2005). This procedural knowledge is used to analyse issues or social phenomena, for example, considering the question of cause and effect (see Sandahl, 2013).

Thereafter, an evaluation was made to determine whether the questions in these sections focused on defined and specific knowledge, or whether they were elaborative, requiring reasoning or an independent evaluation of multiple factors. In the first case, defined knowledge refers to subject-specific knowledge that has a strong framing (Hammarlund, 2012; Persson, 2017). The questions to be asked and the expected answers are predefined. For example, this could involve a question in a textbook where a specific piece of information is required (see Berg, 2021; Persson, 2019). An example of this would be specifying the timing of an event. In the second case, elaborative knowledge is less constrained by a specific domain of knowledge. Instead, it involves the student primarily conducting their own analyses, where no specific answer is required (see Persson, 2019). Rather, the question is open, allowing the student to assess what might be the explanation for the development of an event. This could involve weighing which causes are considered more important than others in explaining a historical event, and then justifying the response (see Berg, 2021).

Based on this, the lesson assignments were classified into three different patterns. Lesson assignments with a student-oriented input were listed alongside the corresponding epistemological outputs, as can be seen in Table 2. Additionally, it was noted whether the assignment had a defined or elaborative knowledge focus. Similarly, assignments were categorised for both context-oriented and analysis-oriented knowledge focus in the entry of the lesson assignments.

These three patterns of the relationship between entry and exit in the lesson assignments are presented as three sections in the results part of the article. They are named 'From elaborative student-oriented to analysis-oriented knowledge', 'From context-oriented to analysis-oriented knowledge' and 'From analysis-oriented to analytical knowledge'. The movement, or epistemic rhythm, that was most common for each section has been highlighted with a blue background, as shown in Table 2. This variation is predominantly described in each section. Where other common variations occur, they are also described as secondary findings. This analysis is presented numerically within each category in Tables 3–5. It serves as an example of how this may appear in the category where student-oriented knowledge forms the starting point for the lesson task.

The final analysis involved an inductive search to identify the didactic function observable in the examples that had been classified under the existing categories outlined previously (see Berg and Persson, 2023). Here, *inductive* refers to a qualitative judgement of the didactic function present within each respective epistemic rhythm. In this case, no predefined criteria were applied; instead, the assessment was based on the orientation of the epistemic rhythm itself. The assumption is therefore that the movement observed as an epistemic rhythm in the lesson assignments is likely to contribute to various didactic functions (see Berg and Persson, 2023). Based on the categories formulated in the results, three distinct didactic functions were identified, referred to as *broaden*, *reorganise* and *evaluate*. The empirical foundation for the analyses conducted will be presented in the results section of the article. This will be done using quotations from the lesson assignments, which thus serve as representative empirical examples of the analysis that was carried out.

Table 2. Example table of analytical categorisation

Entry	Defined	Elaborative	Exit	Defined	Elaborative	Total
Student-oriented	XX	XX	Analysis-oriented	XX	XX	XX
	XX	XX	Context-oriented	XX	XX	XX
	XX	XX	Student-oriented	XX	XX	XX
Total	XX	XX		XX	XX	XX

Results

The analysis of the lesson assignments identified three prominent and different epistemic rhythms, each corresponding to different didactic functions. These epistemic rhythms are presented in the results section, together with a review of the empirical and theoretical foundation for these results.

From elaborative student-oriented to analysis-oriented knowledge

The first, most prominent epistemic rhythm identified among the lesson assignments, highlighted in blue in Table 3, begins with a student-oriented focus at the entry point of the assignment. This is also the epistemic rhythm that occurs the least frequently among the lesson assignments, with 17 examples in the empirical data. Among these, a significant majority – 16 out of 17 – are assessed as having an elaborative focus at the entry point. In contrast, the exit points show greater variation. Of the 17 examples, 12 have an analytical exit and five are focused on student-oriented questions. In both cases, the elaborative focus is dominant, with 15 out of 17 examples.

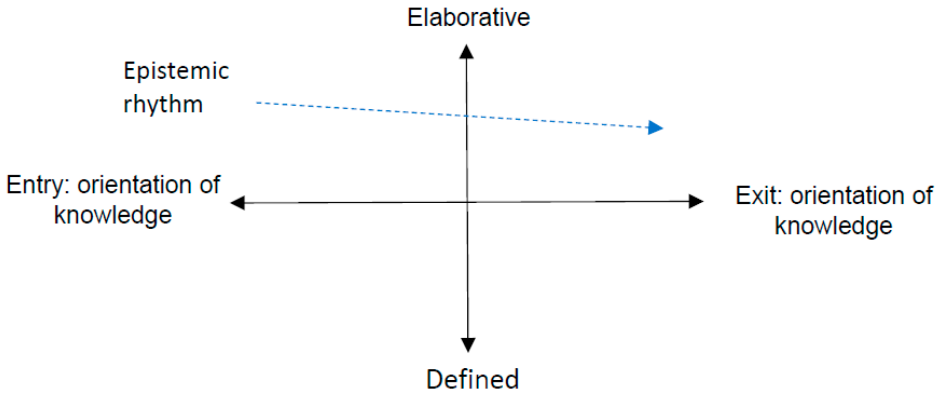
Table 3. Epistemic rhythm with student-oriented entry

Entry	Defined	Elaborative	Exit	Defined	Elaborative	Total
Student-oriented		12	Analysis-oriented	2	10	12
			Context-oriented			
	1	4	Student-oriented		5	5
Total	1	16		2	15	17

The predominant pattern in cases where the entry point of the lesson assignments has a student-oriented knowledge focus is that they contain elaborative questions. The assignment then exhibits a progression towards an analytical orientation at the exit point. This results in an epistemic rhythm, denoted by the dashed arrow (see Figure 2), which is elaborative at both ends but exhibits a slight downward trend, with two defined examples.

Characteristic of student-oriented knowledge is that students are involved in the assignment as subjects in various ways. Students are asked to provide their own perspectives or to imagine something in relation to the questions posed by the lesson assignment. A defining feature here is that the question is elaborative, meaning it does not have a specific or predetermined answer. For example, students might be expected to compare different aspects of something or to make a valuation of various aspects. An example of this is a lesson assignment where students are asked to imagine themselves as another person in a different context, such as a historical figure or a person in a leadership position (L2, 5).

Figure 2. Epistemic rhythm from student-oriented to analysis-oriented knowledge



In other cases, the assignment involves the student reflecting on their own experiences. The example in Box 1 is the predominant pattern in this category. It is an assignment that involves questions about online anonymity in contemporary society. The assignment also includes a form of contextualisation, where the development is to be viewed in relation to the establishment of democratic principles in the Nordic countries. Additionally, the questions incorporate a future-oriented perspective on developmental trajectories over time. As indicated by the initial six questions, students are asked to make their own evaluations. These evaluations are also elaborative in nature. The assignment involves considering situations where people might want to remain anonymous online, as well as making personal comparisons. There are also questions directed towards the student's own life world. The question, shown here in orange text, reads, '3. When is it important for you to be anonymous online?' Additionally, the personal address is present in the concluding question, which asks what a ban might mean for different people in various contexts, including the question, also shown in orange text, directed towards the student: 'For you personally?'

In the exit point of the lesson assignment, a shift in the orientation can be observed towards analysis-oriented knowledge. In this case, students are expected to reason about societal conditions based on the text and source provided in the link. Unlike the questions tied to the student's life world in the entry, these are linked to various political science issues where online anonymity is connected to questions about democratic rights. The questions are notably analytical, including the example highlighted in blue, with students expected to reason using procedural concepts such as explanation and comparison. There is also an elaborative stance, where it is not apparent what the students are expected to conclude. Emphasis is placed on the ability to investigate the issue. Additionally, the assignment includes a component where the questions are first discussed within the assignment and then in the class.

As shown in Table 3, outcomes often include a student-oriented focus in addition to a more analysis-oriented knowledge focus. In one assignment that had both a student-oriented entry and outcome, the intention is for the students to relate their own lives to societal developments after conducting an initial analysis of five major contemporary events, both nationally and globally. In the next step, an analysis is requested where the student's own life is related to these events. The question reads: 'In the next step, you should relate these five events to yourselves. Have they affected you in any way?' (L2, 5). This includes a question about how the social and cultural context of the student's upbringing has contributed to their identity. There is also a forward-looking aspect where students are asked to consider how socialisation might have influenced, for example, opinions and interests.

The most prominent epistemic rhythm in this category is a movement from student-oriented and elaborative knowledge at the entry point of the lesson assignment to a more analytical and elaborative approach at the conclusion. This transition highlights a didactic function that is assessed as *broadening* the understanding of the thematic content being explored. This means that, starting from an elaborative question connected to the student's own lifeworld, the student broadens their understanding using elaborative and analytical knowledge to address the question. In Box 1, this can be observed when the student is first asked, at the entry point of the assignment, to reflect on what the concept of online

anonymity means to them personally. Then, at the conclusion of the assignment, the student is asked to broaden their understanding by conducting an analytical and elaborative assessment of the relationship between anonymity, freedom of speech and democracy more generally. In this way, there is a movement in the lesson assignment from the student's personal perceptions and experiences of the issue to a broader, more general and elaborative analytical level.

Box 1. Discuss questions about online anonymity in pairs (Source: Lesson task from Teacher 10)

1. What different situations where people might want to be anonymous can you think of? Make a list.
2. Is it important to be able to remain anonymous online? Provide reasons and highlight several arguments for and against.
3. When is it important for you to be anonymous online?
4. Do you think there is a difference in how we express ourselves when writing under an anonymous alias? Do people behave differently?
5. What are the advantages and disadvantages of people being able to be anonymous? Give some examples and explain your reasoning.
6. What would a potential ban on anonymity mean? For you personally? For others? For citizens in a democracy? For people in a dictatorship?

We will discuss these questions as a whole class, so be prepared to participate in the discussion.
Read the text and reflect on the pros and cons of anonymity (in pairs or individually)
Excerpt from: Hate and threats online
One distinguishing feature of communication on the internet is that it can happen anonymously, or at least there is a perceived anonymity, as it is possible to hide behind a fictitious identity (Dunkels, 2016). Anonymity has been considered an important component of participating in democratic discussions. This is particularly true for certain vulnerable groups, both in countries with undemocratic regimes and in democracies such as the Nordic countries, where certain groups are particularly vulnerable and therefore need anonymity. However, anonymity also has a downside, namely that those who engage in online hate do not have to, or do not believe they have to, confront the responsibility that follows. Perpetrators online may become harder and, in some cases, impossible to prosecute for crimes. There are two types of anonymity: users' anonymity in relation to other users, and anonymity in relation to the service provider. The latter anonymity is, however, limited because certain information is required for registration and IP addresses are logged (Dunkels, 2016; Sackemark et al., 2015).
Source: Hate and threats online (PDF)
Is it a democratic right to remain anonymous?
In which situations is anonymity needed?
Is there a connection between anonymity, freedom of expression and democracy? Explain!
Can you be anonymous online?
Do you think people will be able to remain anonymous on the internet of the future? Discuss!
We will discuss these questions as a whole class, so be prepared to participate in the discussion.

From context-oriented to analysis-oriented knowledge

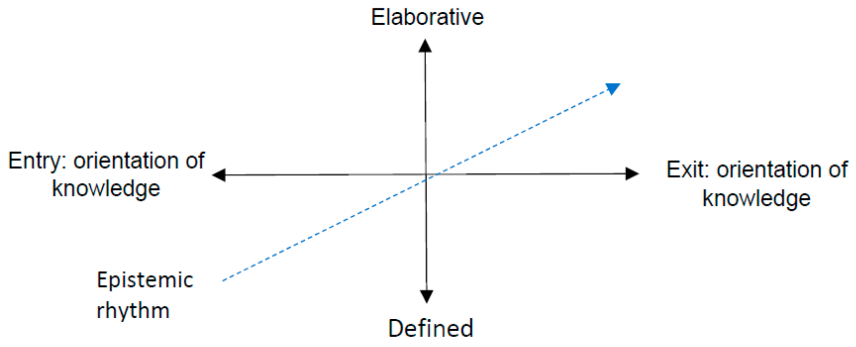
The second most prevalent epistemic rhythm among the lesson assignments, highlighted in blue in Table 4, is one that has a context-oriented knowledge focus at the entry point. In total, there are 44 lesson assignments in the empirical material with this entry focus. It is also noteworthy that 42 of these 44 lesson assignments have a defined knowledge focus in their questions. When considering what is requested in the outcome of these lesson assignments, analysis-oriented knowledge predominates, with 38 examples. In these cases, elaborative knowledge in the questions, with 27 examples, is the most common, while defined knowledge corresponds to 11 examples.

The main pattern in the lesson assignments with a context-oriented focus at the entry point is that the knowledge is defined. There is then a movement towards analytical knowledge, which is identified as laboratorial. This results in an epistemic movement that follows a diagonal trajectory, as illustrated in Figure 3. This epistemic rhythm thus differs significantly from the one where student-oriented knowledge is sought at the entry point.

Table 4. Epistemic rhythm with context-oriented entry

Entry	Defined	Elaborative	Exit	Defined	Elaborative	Total
Context-oriented	1		Student-oriented		1	1
	36	2	Analysis-oriented	11	27	38
	5		Context-oriented	5		5
Total	42	2		16	28	44

Figure 3. Epistemic rhythm from context-oriented to analysis-oriented knowledge



One example where this epistemic rhythm is prominent is the lesson assignment described in Box 2, where the form of the assignment is to record a podcast. The instructions emphasise the importance of addressing questions of cause and effect related to historical revolutions. However, at the entry point to the assignment, there is a preparatory phase where the textbook and study questions provide contextual knowledge for the work on the podcast. As is evident from the questions that are to be processed in relation to the podcast script, the focus is on defined knowledge, where specific answers are sought, for instance, the question highlighted with orange text: 'Which of the revolutions was most important for the development in the area/theme you have chosen?'

In the concluding phase of the assignment, there is a marked shift towards a more analysis-oriented approach to knowledge. This stage involves summarising a broader historical event while considering potential future scenarios. It culminates in an evaluation of possible future outcomes. This analytical focus is reflected in the questions highlighted in blue text: 'What has changed in the chosen area from the revolutions at the end of the 18th century until today? Conclude with a discussion: what possible future scenarios do you foresee? What do you think will happen in the future?' In this section, the emphasis transitions from definitional to elaborative knowledge, requiring analysis and reasoning that leads to a possible or plausible conclusion.

As can be observed, it is also common in the exit phase of this category for defined questions to appear. For example, in a lesson assignment where students are required to map various criminal cases and subsequently record a podcast, the questions demand specific aspects of analytical knowledge. The questions seek specific answers, such as: 'The nature of the crime and events. 2/ Important person(s) involved. 3/ What were the verdicts? Appeal?' (L8, 8). Here, precise answers are required regarding who was involved in the case and what the verdicts were.

Overall, the most prominent epistemic rhythm in this category moves from a defined, context-oriented knowledge to a more elaborative and analysis-oriented knowledge. The epistemic rhythm resulting from this movement is assessed as *reorganising* previously given knowledge to contribute to understanding a new question. This entails using a defined field of reference knowledge as a starting point to conduct an elaborative analysis where the knowledge is reorganised to provide explanations, for example, of an event. In the podcast example, this involves starting from defined contextual knowledge about the causes of the French Revolution and performing an elaborative analysis to reorganise that knowledge in order to understand the consequences of the event up to the present day. Through this

movement from reference-oriented knowledge, students are thus required to reorganise the knowledge to create understanding in a new context.

Box 2. Instructions for the Podcast (Source: Lesson task from Teacher 5)



You will record a podcast where you discuss a specific theme/area (found in a separate document). The theme must be connected to the French and/or Industrial Revolution. When conducting your research, follow the checklist below and keep the discussion questions in mind at all times. Historically, the task is not time-bound: you can choose to discuss consequences extending into the 20th century. The important thing is to find relevant connections (cause and effect) between the revolution and the theme you've chosen.
— The length of the podcast should be 3 to 5 minutes per student, and each group should consist of 3 students. Thus, each podcast will be approximately 10–15 minutes long.
Questions:
1/ How have the French and Industrial Revolutions influenced the world in the area or theme you have chosen? What did the world look like before these revolutions, and what changes did they bring about? In what ways can the area or theme you have chosen be seen as a consequence of the Industrial and French Revolutions? Which of the revolutions was most important for the development in the area/theme you have chosen?
2/ Are there any individuals who can be considered particularly important for the development of the area or theme you are studying? Discuss how the subject can be analysed from both an actor and process perspective.
3/ You have studied, analysed, and described the development over many years. Summarise the progress within your area or theme from the 1700s to today. What has changed in the chosen area from the revolutions at the end of the 18th century until today? Conclude with a discussion: what possible future scenarios do you foresee? What do you think will happen in the future?
Important! Remember to use various perspectives when answering the discussion questions. These could include KEPS (Cultural, Economic, Political, and Social perspectives), Materialism and Idealism, Actor and Process, or other relevant perspectives that suit your theme.

From analysis-oriented to analytical knowledge

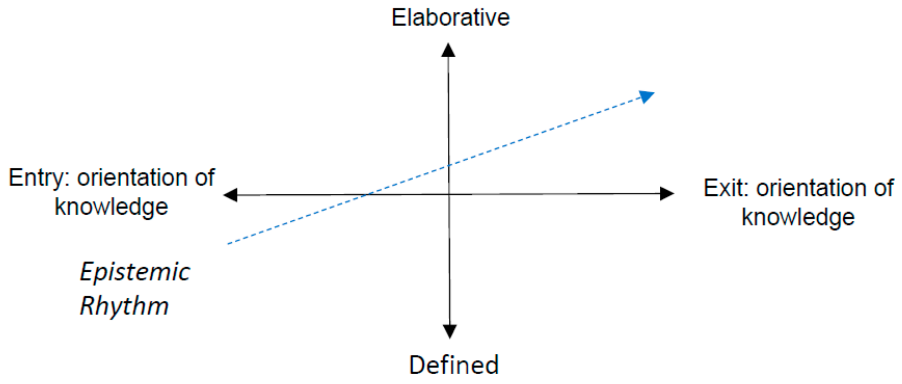
The final epistemic rhythm originates from an analysis-oriented entry point in the lesson assignments. A total of 53 lesson assignments have this entry point. In these cases, the most common exit is also focused on analysis-oriented knowledge. Here, there are 45 examples, representing a substantial majority. While the majority of lesson assignments are defined in the entry phase, with 26 examples, a relatively significant number, 21 examples, are also elaborative. In terms of the exit phase, there is instead an overarching majority, with 41 lesson assignments being elaborative (see Table 5).

Table 5. Epistemic rhythm with analysis-oriented entry

Entry	Defined	Elaborative	Exit	Defined	Elaborative	Total
Analysis-oriented		1	Context-oriented		1	1
	6	1	Student-oriented	1	6	7
	26	19	Analysis-oriented	4	41	45
Total	31	21		5	48	53

The main pattern that emerges in assignments where the entry is characterised by analysis-oriented knowledge is that, predominantly, the knowledge is defined. In the assignment, there is then a movement towards an exit that is characterised by analytical and elaborative knowledge orientation. As illustrated in Figure 4, the blue dashed arrow represents a different epistemic rhythm.

Figure 4. Epistemic rhythm from analysis-oriented to analytical knowledge



It is most common for the culmination of these assignments to focus on analysis and defined knowledge at the entry point and analytical and elaborative knowledge at the exit point of the assignment. An example can be seen in Box 3 of a classroom assignment where students are asked to compare Sweden's political system and the issue of human rights with another country. In the assignment, they first conducted an analysis of Sweden based on the questions provided. They were then instructed to work in pairs to analyse another country using the same set of questions. The assignment's questions span from a contemporary analysis of current conditions, which are to be examined against earlier historical social principles such as Montesquieu's separation of powers, to the background of the contemporary political landscape today.

The initial questions, shown in the text highlighted in orange, can be characterised as analytical in nature, as they require students to evaluate the structure of the political system. However, this evaluation must be conducted using predefined categories such as 'monarchy, republic, empire' and 'king/queen, president, sultan, emperor, etc.' The knowledge being sought is therefore analytical but also structured within clearly defined parameters.

Box 3. Democracy, Political Systems, and Human Rights (Source: Lesson task from Teacher 6)

Purpose of the Assignment
The purpose of this assignment is to apply the concepts of democracy, human rights, and Sweden's political system to the study of political systems in different countries around the world.
Description of the Assignment
To begin, you have gained insight into Sweden's political system and parliament. Now, you will work in pairs. Each pair will select a country to examine. You will investigate the political system of the chosen country and analyse how it adheres to human rights principles. Base your work on the following points:
Questions about the Political System and Head of State
What type of political system does the country have (monarchy, republic, empire, etc.)? What type of head of state does the country have (king/queen, president, sultan, emperor, etc.)?
Questions Regarding the Government (the Executive Branch)
How is the government formed in the country? Does the head of state or head of government play an active role in the formation of the government? What powers does the government have in the country? What traces of Montesquieu's principle of the separation of powers can be observed in the country? Is the country characterised by parliamentarism or presidentialism?
Questions about Human Rights
Develop a thorough understanding of the state of human rights in the country. Which rights are respected, and more importantly, which are not? Select 1–3 examples of human rights that are particularly notable in the country. What might explain why the situation is as it is?

In the concluding question, where the text is highlighted in blue, students are asked to provide a final evaluation of human rights in the selected country. Here, the analysis is expected to be conducted using an open and self-selected set of criteria to assess how well the country adheres to the principles

of human rights. There is also an exit point where the students are required to explain the situation using a retrospective perspective. In relation to this, there are also historical questions that require the students to provide an analysis. 'What traces of Montesquieu's principle of the separation of powers can be observed in the country?' As indicated in the assignment, students are also expected to compare the countries based on the information they have analysed. Thus, the culmination of the assignment exhibits a distinctly analytical and elaborative character, reflecting the type of knowledge being sought.

This category also includes examples where the exit has a student-oriented knowledge focus. For instance, a lesson assignment in religious studies involves the analysis of ethical issues, such as social and relational problems, for example, if a friend drinks too much alcohol. In the entry phase of the assignment, the questions are directed towards specific answers. The assignment is to clarify who is involved in the situation. The questions are: 'A. Who is affected by the problem? B. What action alternatives are available? C. What are the consequences of the different action alternatives for those involved? D. What ethical principles, norms, and/or other values can be applied?' The questions, as can be seen, also have an analytical and procedural focus. In the exit phase, the questions take on a more elaborative character, directed towards the student's judgement. The student is asked to provide their own reasoning for the advice they should give to their friend. The question is: 'Explain why you gave the advice you did, where you reason and provide examples of whether you used consequentialism, deontology, or virtue ethics in your advice' (L6, 4). Additionally, an analysis is requested regarding how the advice given by the student reflects different ethical theories.

In this category, the most prominent epistemic movement is from analytical and defined knowledge to analytical and elaborative knowledge in the lesson assignments. Within this epistemic rhythm, the didactic function is contributing to the student's ability to *evaluate* various forms of knowledge. It refers to movement from analysing a predefined set of factors to developing the ability to analyse and evaluate a non-predetermined set of factors. For example, as discussed earlier, this could involve making a judgement about which criteria should be applied in the analysis, in contrast to the initial phase of the task where the criteria were pre-established. The didactic function of this epistemic rhythm is thus assessed as enabling the evaluation of prior knowledge. By transitioning from more defined analytical knowledge to elaborative analytical knowledge, the student's ability to evaluate different aspects of knowledge in relation to one another is enhanced.

Summary and conclusions

This article aims to investigate the didactic functions that can be identified in school lesson assignments. Two research questions have been posed to fulfil this aim: What epistemological positions can be identified in school lesson assignments? What epistemic rhythms between different epistemological positions can be identified in school lesson assignments?

Before discussing the research questions, I would like to shed some light on the empirical foundation of the article. It should be noted that the study comprises 124 lesson assignments from 10 Swedish teachers. This sample is somewhat limited in scope, making it challenging to draw more generalised conclusions. At the same time, it represents a finite selection of lesson assignments and serves as an illustration of the types of tasks used in Swedish schools. These are thus cases occurring within a Swedish school context, among potentially many more (see Yin, 2009, pp. 3–7). It is also essential to emphasise that the analysis does not focus on individual teachers' lesson assignments or specific subjects. Rather, the lesson assignments have been regarded as representations of school subjects addressing historical and contemporary societal issues. The primary interest has been in the assignments as a collective whole (see Berg and Stolare, 2024). The categories identified should therefore be seen not as exhaustive but as indicative of this specific empirical dataset. However, the initial phase of analysis was conducted with reference to categories of epistemological positions noted in previous empirical studies (see Berg, 2014; Evans, 1994). This study is thus to be regarded as an early step in exploring these issues, and further research is needed to contribute additional insights to this field (see Schüllerqvist and Osbeck, 2009).

When focusing on the first research question, this article has identified various epistemological positions in the entry and exit points of lesson assignments. This indicates that the assignments themselves target knowledge of different types. In this case, student-oriented, reference-oriented and analysis-oriented knowledge has been used as a theoretical frame (see Berg, 2014). It has also been

shown, as confirmed by previous studies, that the knowledge requested can, to varying degrees, be well defined or have a more elaborative focus. That is, on the one hand, something specific is requested that can be answered with knowledge presented in a textbook, while on the other hand, the student is asked, for example, to take a stance where different causes of an event must be evaluated (see Berg, 2021; Persson, 2019).

What has been identified is that these shifting knowledge focuses become prominent to varying extents within the same lesson assignment. Additionally, a movement can be observed where a certain type of knowledge is requested at the entry point of the assignment and another type of knowledge at the exit point. In this way, it has been a matter not only of different aspects of knowledge being requested but also of a shift from one knowledge focus to another within the lesson assignments. This movement, which has been noted in several previous studies regarding other practices of history teaching, is referred to in this article as epistemic rhythm (Berg, 2023; Berg and Persson, 2023).

When it comes to research question 2, which focuses on the nature of the movement between different epistemic positions, three epistemic rhythms have been identified in the empirical data, representing the movement between the entry and exit points of the lesson assignment. These are presented in Table 6 alongside their didactic functions. The first epistemic rhythm involves a movement from student-oriented to analysis-oriented knowledge. In both the entry and exit points of the assignment, the knowledge focus is elaborative. The didactic function in this case is assessed as broadening the student's perceptions of the knowledge object. An example of this is when a student is initially asked how they themselves, from their point of view, perceive online anonymity. This then transitions to an analytical conclusion, situating the question within a societal context in terms of cause and effect.

Table 6. Epistemic rhythm and didactic function in lesson assignments

Epistemic rhythm	Entry point	Exit point	Didactic function
From student-oriented to analysis-oriented knowledge	Elaborative	Elaborative	Broaden
From context-oriented to analysis-oriented knowledge	Defined	Elaborative	Reorganise
From analysis-oriented to analysis-oriented knowledge	Defined	Elaborative	Evaluate

The second prominent epistemic rhythm involves a movement from context-oriented to analysis-oriented knowledge. Here, the movement is characterised by knowledge being defined at the entry point and elaborative at the exit point. The conclusion is that this epistemic rhythm contributes to reorganising prior knowledge. This is exemplified in a lesson assignment where students are initially asked to provide specific facts about a historical revolution, progressing to a concluding question that requires them to discuss how these revolutions have influenced contemporary society.

Finally, the third epistemic rhythm involves a movement where both the entry and exit points have an analytical knowledge focus. The distinction lies in the entry point requiring defined knowledge, while the exit point emphasises an elaborative knowledge focus. In this case, the assessment suggests that this rhythm contributes to an evaluative didactic function. An example of this is a lesson assignment where students are initially asked to provide precise factual knowledge about a country's political system, progressing to questions at the conclusion where they are expected to analyse how well the country aligns with human rights criteria. These different selections, in this case referred to as epistemic rhythm, contribute to various didactic functions and, in many ways, confirm the findings of previous studies (Berg and Persson, 2023; Persson, 2022).

This article confirms several findings of previous research. At the most general level, the selection of knowledge content can be linked to different knowledge domains (for an overview of research on this matter, see Elmersjö and Zanazanian, 2024). This has been criticised by some researchers, who argue that teachers, in their selection of knowledge, operate in relation to varying epistemic positions in ways that are inconsistent (Miguel-Revilla, 2022; Stoel et al., 2022). One concept used by some researchers to describe this phenomenon is that teachers' selection of content *wobbles* in an epistemic manner. The overall conclusion drawn by researchers from this perspective is that teachers appear to lack awareness

of epistemic positioning when selecting subject content (see Wagner, 2024). These analyses are often conducted using theoretical perspectives grounded in defined categories. In some cases, for example, these categories and concepts are drawn from traditions in history didactics research, while in others they originate from more general knowledge-theoretical domains (Kuhn et al., 2000; Lee, 2005; Lee and Ashby, 2000; Lee and Shemilt, 2003; Schommer, 1990).

Another complementary way to understand teachers' movement between different knowledge positions is to view it as part of a broader spectrum within an educational setting – not only in relation to specific subjects but also as stemming from the school's civic educational mission and playing a crucial role in students' meaning-making (Berg and Persson, 2023; Deldén, 2023). Other researchers have also addressed this broader perspective and called for studies with such a focus, particularly on the everyday work of selecting knowledge for teaching. They argue that it can be difficult to distinguish between teachers' ideal views on knowledge and their work within the broader educational framework (Maggioni et al., 2009; see also Elmersjö and Zanazanian, 2024; Stoel et al., 2022). In this context, Ledman (2014) has also pointed to the weak classification and framing of, for instance, history education as a school subject. Parks also suggests that this epistemic fluency must be seen in relation to the teachers' historical-cultural environment and ideas on knowledge formulated in that context (Parkes, 2024; see also Berg, 2014). Altogether, this highlights the importance of also taking these perspectives into consideration when trying to understand teachers' selection and formulation of knowledge for teaching and learning.

By adopting a different theoretical framework, as is done in this article, referred to as the perspective of the teaching profession (see Berg, 2021; Berg and Persson, 2023; Persson, 2022; Persson and Berg, 2022), an alternative logic underpinning the selection of knowledge can be discerned (see Grossman and McDonald, 2008; Loughran, 2019). Within this theoretical framework, the movement between different positions can possibly be seen to address situated conditions and diverse educational goals in education. Alternatively, this movement may be interpreted as a means of linking various aspects of knowledge to achieve distinct didactic functions, thereby addressing different epistemological ideals. In this way, shifts in knowledge focus within lesson assignments can be understood as deliberate choices aimed at fostering a diversity of educational goals that align with the realities of teachers' professional practice. The logic behind the selection can, therefore, potentially be seen as a practice-oriented theory where, instead of being grounded in general, predefined theories of knowledge, it is shaped by the situated conditions of the specific teaching context. In such a case, the selection of knowledge – and, as demonstrated in this article, the formation of the epistemic rhythm within lesson assignments – appears logical, as it is anchored in a contextual framework grounded in the teacher's practice-oriented theory, developed over time in response to the school's situated conditions and its interpretation of the subject's disciplinary orientation (Berg, 2014; Berg and Persson, 2023; Loughran, 2019; Persson, 2022).

Let us summarise and draw conclusions from these results. First, the findings highlight the importance of considering movement within a teaching sequence to understand the teaching and learning ambitions as well as the possible outcomes (see Berg and Persson, 2023; Persson, 2022). Rather than static depictions of test construction or observations of classroom activities at specific locations, the focus should be on movement within sequencing. Here, the relationship between knowledge contents in a lesson assignment has demonstrated the importance of understanding how different forms of knowledge can be interconnected in what has been termed epistemic rhythm (see Berg, 2023).

Second, it can be argued that the movement between different aspects of knowledge within a lesson assignment demonstrates that the concept of epistemic rhythm can be a powerful tool for making visible both the movement within a teaching and learning activity and the didactic functions that can be interpreted in the selection of knowledge in a lesson assignment (see Högberg, 2015). In this way, the concept can be used to discuss how different movements within a lesson assignment, an examination or a lesson plan contribute to various educational goals.

Third, the concept has the potential to make visible a practice-oriented logic, referred to by researchers as practical wisdom, for the selection of knowledge that considers a broader view of the teacher's role (Frelin, 2010; Loughran, 2019). Instead of viewing the movement between different aspects of knowledge as an inconsistency, it can be seen as part of the teacher's work in meeting various educational goals. This should be viewed in the context that teachers must consider, among other things, the changing formulations of the curriculum, students' prior understanding and the concept of progression in teaching (Berg, 2014; Grossman and McDonald, 2008).

Fourth, these results highlight the need for studies that examine subject-didactic choices in teaching from multiple theoretical perspectives. Using the academic subject's theories of knowledge

and/or general perspectives on knowledge is important, but it must be complemented by studies that adopt both a broader understanding of teachers' professional responsibilities and theoretical perspectives that seek to understand in greater depth the background of knowledge selection (see Berg and Persson, 2020; Elmersjö and Zanazanian, 2024; Grossman and McDonald, 2008; Loughran, 2019; Persson and Berg, 2024; Thorp and Persson, 2020).

Data and materials availability statement

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations and conflicts of interest

Research ethics statement

The author conducted the research reported in this article in accordance with the research ethics standards of Dalarna University.

Consent for publication statement

The author declares that research participants' informed consent to publication of findings – including photos, videos and any personal or identifiable information – was secured prior to publication.

Conflicts of interest statement

The author declares no conflicts of interest with this work. All efforts to sufficiently anonymise the author during peer review of this article have been made. The author declares no further conflicts with this article.

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